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special communication

Isolation of interstitial fluid from rat mammary tumors by a centrifugation method

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Wiig, Helge, Knut Aukland, and Olav Tenstad. Isolation of interstitial fluid from rat mammary tumors by a centrifugation method. *Am J Physiol Heart Circ Physiol* 284: H416–H424, 2003. First published September 12, 2002; 10.1152/ajpheart.00327.2002.—Access to interstitial fluid is of fundamental importance to understand tumor transcapillary fluid balance, including the distribution of probes and therapeutic agents. Tumors were induced by gavage of 9,10-dimethyl-1,2-benzanthracene to rats, and fluid was isolated after anesthesia by exposing tissue to consecutive centrifugations from 27 to 6,800 *g*. The observed ^{51}Cr-EDTA (extracellular tracer) tissue fluid-to-plasma ratio obtained from whole tumor or from superficial tumor tissue by centrifugation at 27–424 *g* was not significantly different from 1.0 (0.92–0.99), suggesting an extracellular origin only. However, fluid collected from excised central tumor parts had a significantly lower ratio (0.66–0.77) for all imposed G forces, suggesting dilution by fluid deriving from a space unavailable for ^{51}Cr-EDTA. The colloid osmotic pressure in tumor fluid was generally higher than in fluid isolated from the subcutis, attributable to less selective capillaries and impaired lymphatic drainage in tumors. HPLC analysis of tumor fluid showed that low-molecular-weight macromolecules not present in arterial plasma were present in tumor fluid obtained by centrifugation and in venous blood draining the tumor, most likely representing proteins derived from tumor cells. We conclude that low-speed centrifugation may be a simple and reliable method to isolate interstitial fluid from tumors.

extracellular fluid; colloid osmotic pressure; collagen; glycosaminoglycans

ANALYSIS OF INTERSTITIAL FLUID, i.e., the fluid bathing the cells in a tissue, may give significant information on fluid exchange. Proteins dissolved in interstitial fluid will exert a colloid osmotic pressure (COP) opposing the corresponding pressure in plasma and will most likely influence the interstitial fluid pressure. Interstitial fluid is, however, not readily accessible in the normally hydrated state, and therefore various techniques have been developed for tissue fluid sampling (for review see Ref. 2).

Access to interstitial fluid in solid tumors is of special interest because this may have potential therapeutic consequences. As shown by us and several others (26) (for review see Ref. 9), tumors have a high interstitial fluid pressure, which may influence drug delivery (13). One of the determinants of interstitial fluid pressure is the COP of interstitial fluid, again depending on the protein concentration of interstitial fluid. Therefore, access to interstitial fluid may be of fundamental importance to understand transcapillary passage of substances from vessels to interstitium. Furthermore, the emergence of immunoglobulins in solid tumor therapy makes it of interest to measure the concentration and distribution volume of macromolecules as well as other substances in the fluid bathing tumor cells.

Despite the interest, no interstitial fluid sampling method is generally accepted. This may partly be due to the nature of solid tumors. In other tissues, lymph has been accepted as a measure of interstitial fluid (2). Although intratumoral lymph vessels have been observed (11, 18), these have been shown to be nonfunctional (11), making lymph sampling inapplicable in this tissue. In a classical study, Gullino and co-workers (8) sampled interstitial fluid from porous chambers implanted in tumors. Another approach was used by Sylven and Bois (21) who collected fluid from a sectioned surface by using micropipettes, risking the contamination from cellular fluid. In tissues other than solid tumors, wicks have been used for interstitial fluid sampling (2), and in a recent study Stohrer and co-workers (19) sampled fluid from solid tumors by using nylon wicks. Wick implantation in richly vascularized tumor tissue may result in cell and vessel damage leading to contamination of the isolated fluid by intracellular fluid and blood, and therefore alternative methods are recommended. Communication with in-

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terstitial fluid may also be obtained by using the microdialysis technique (7), but this is a technique suitable for the study of tumor distribution of small molecules, and again the probe insertion may result in vascular damage.

In a series of studies, Aukland and co-workers (1, 3, 22) have isolated tissue fluid from the cornea and rat tail tendon by centrifugation. These tissues have a low cell content and are rich in collagen, which may restrict the removal of macromolecules when high G force is applied. Tumor tissue has at least an order of magnitude higher hydraulic conductivity than cornea (9, 20), suggesting that tissue fluid could be isolated from tumor tissue by centrifugation. We evaluated centrifugation as a method for tumor fluid isolation in a chemically induced mammary carcinoma and found that fluid isolated this way may be representative for tumor interstitial fluid. This new method may therefore be useful when studying interstitial transcapillary fluid balance and interstitial distribution of probes, including therapeutic agents.

MATERIALS AND METHODS

Experimental animals. The experiments were performed in anesthetized female Sprague-Dawley rats, 207–260 g at the time of the experiment, fed a standard laboratory diet. At 7 wk of age, the rats were given 16 μg 9,10-dimethyl-1,2-benzanthracene in olive oil by gavage in the animal facility at M&B, Ry, Denmark. After gavage the rats were housed in isolation for 1 wk and subsequently transferred to our animal facility. The rats were housed under standard animal housing conditions until mammary tumors developed along the mammary crest 6 to 8 wk later. All experiments were performed in accordance with recommendations given by the Norwegian State Commission for Laboratory Animals and were approved by the local ethical committee.

The rats were not fasted before or during the experiments. At the day of the experiment, the rat was anesthetized with pentobarbital (50 mg/kg ip). While the rat was anesthetized, body temperature was maintained at 36.5–37.5°C with a heating lamp. Unless otherwise specified, blood was sampled by cardiac puncture, and the rat was killed by intracardiac injection of saturated potassium chloride.

Centrifugation technique. The anesthetized rat was immediately transferred to an incubator kept at room temperature (20–24°C) and 100% relative humidity. Tumors were excised, flushed with saline to remove blood from the surface, blotted gently with tissue paper to remove excess saline, and transferred to 2-ml centrifuge tubes used for isolation of tumor fluid. Tumors weighing 0.25–1.0 g were transferred intact to the tube, and results from these are later referred to as intact tumor. Larger tumors (>1 g) were sectioned, and the tumor sections (~0.5 g) were placed with the intact (i.e., unsectioned) surface facing downwards toward the nylon mesh, later referred to as tumor pole (see RESULTS). In some tumors, central parts (also ~0.5 g) were excised for centrifugation avoiding obviously necrotic areas.

The preweighed centrifuge tubes were provided with a basket of nylon mesh with pore size ~15 × 20 μm designed to keep the sample up from the bottom of the tube (1). The tube was immediately capped, reweighed, and spun in an Eppendorff 5417 R centrifuge placed in a cold room at 4°C, and then immediately brought back into the incubator. The basket containing the sample was transferred to another preweighed centrifuge tube, and the tumor fluid accumulated at the bottom was collected in glass capillaries that were either closed immediately for later analysis or diluted in buffer for HPLC. After being reweighed, the second tube was centrifuged, and the procedure was repeated by centrifugation of the basket in a third, fourth, and sometimes fifth tube. As described previously (1), the protocol provided data on the initial and subsequent weights of the tumor sample, each centrifugation volume, as well as any fluid loss by evaporation. Typical tumor fluid samples were 1–15 μl.

To reduce the risk of cell compression, the centrifugation speed was lower than in the experiments with the tail tendon. In initial experiments, the samples were spun at 500 rpm (27 *g*) for 10 min, and the speed was increased successively in 100-rpm steps until a sample appeared at the bottom of the tube. This usually did not occur until a speed of 800 rpm (68 *g*) was used, so in later experiments centrifugation speeds of 800, 1,000 (106 *g*), 2,000 (424 *g*), 4,000 (1,700 *g*), and 8,000 rpm (6,800 *g*) were used.

Fluid isolated from tumors by centrifugation was compared with fluid obtained from the subcutis isolated by centrifugation or isolated from dry wicks implanted postmortem. About 0.5 g of skin was excised and placed with the subcutis facing the net, allowing fluid to be collected at the bottom of the tube. Dry wicks were inserted postmortem in the back subcutis and hindlimb skin as described in a previous publication (24). After a 20-min implantation period, the wick ends along with any blood-stained portions were cut off, and the remaining sections were transferred to glass vials filled with 1 ml 0.02% azide saline for elution or were transferred to oil-filled centrifugation tubes provided with a funnel for isolation of native wick fluid (24).

Analyses of tumor fluid. Our aim was to isolate fluid from the tumor interstitium. To estimate any contribution of cellular fluid, we therefore studied the concentration ratio of an extracellular tracer in tumor fluid and plasma. After anesthesia and placement of a polyethylene-50 catheter in a jugular vein, both kidney pedicles were ligated via flank incisions, and 60–70 μCi ^{51}Cr-EDTA were injected intravenously for distribution in the interstitial fluid and measurement of extracellular fluid volume. After 115 min of tracer equilibration, 3–4 μCi of ^{125}I-labeled human serum albumin were injected intravenously and allowed 4 min of equilibration. Blood was sampled by cardiac puncture for isolation of plasma, and the rat was killed and transferred to the incubator. Tumors were removed and transferred to centrifuge tubes and were handled as described above. Tumor fluid isolated in these experiments was pipetted into 1 ml of saline contained in counting vials. Tissue samples taken from the tumor and back skin for calculation of plasma equivalent distribution volumes (in counts$\cdot$min$^{-1}\cdot$g tissue^{-1}/counts$\cdot$min$^{-1}\cdot$ml plasma^{-1}) were placed in tared, covered vials and weighed.

Samples were counted in a LKB gamma counter (model 1282 Compugamma) using window settings of 530–690 keV for ^{51}Cr and 120–320 keV for ^{125}I. Standards were counted in every experiment to obtain spillover corrections, and counts were corrected for background and spillover.

To investigate whether the ^{51}Cr-EDTA was bound to tumor tissue (i.e., not free in the interstitial fluid), tumors that had been equilibrated with ^{51}Cr-EDTA were eluted in saline. After tracer equilibration, tumors were excised and finely minced, and the minced tissue was placed in a tube and counted. After counting was completed, the tissue was suspended in 10 ml PBS containing 0.02% azide by vigorous shaking and left for 24 h in an agitator at room temperature. After centrifugation and removal of as much supernatant as possible, a new aliquot of azide saline was added and the procedure repeated. The counts remaining in each individual tissue sample after elution for 48 h divided by the initial total counts of the same sample, with correction for isotope decay occurring during the extraction procedure, should be an indicator for tissue binding of tracer.

Colloid osmotic pressure. The COP in isolated fluid from tumors and in serum was measured in a colloid osmometer designed for submicroliter samples (23), using membranes with cut-off size of 10 kDa.

Characterization of fluid isolated from tumors and subcutis. The distribution of macromolecules in the tumor fluid isolated by centrifugation, elution fluid of intact or homogenized tumors (see RESULTS), wick fluid, and serum was determined by HPLC using Superdex 75 HR 10/30 or Superose 12 HR 10/30 size exclusion columns (Pharmacia-Biotech) with optimal separation range of 3–70 kDa and 10–300 kDa, respectively. About 1 μl of isolated tumor fluid in a total volume of 100 μl eluent (NaP buffer, 0.15 M, pH 7.4) was injected onto the HPLC system using a Gilson 234 autoinjector (200 μl loop). Constant flow of 1 ml/min was obtained by a Spectreseries P2000 pump (Thermo separation products), and the protein concentration in the elution fluid was measured by UV detection at 280 nm (Spectraseries UV100). The UV signal was digitalized, sampled at 2 Hz, and analyzed on a computer using ChromoQuest (version 2.51, ThermoQuest).

Samples of fluid from tumor, subcutis, and plasma and standard proteins ranging from 2.5–200 kDa in molecular mass (Mark12, Invitrogen) were also analyzed by SDS gel electrophoresis using a vertical mini-gel system (CBS Scientific) and 10% Tris-glycine precast gels (Novex). The gels were run at 125 V for about 90 min and stained with colloidal blue (Novex).

HPLC of fluid isolated from tumors revealed a UV peak around 40 kDa, which was not present in serum. This fraction was subjected to electrophoresis to increase the resolution. During HPLC, a 2-ml sample of eluate containing this peak was collected from three consecutive tumor centrifugates (68, 106, and 424 *g*) of two tumors. The content was pooled and freeze-dried to increase the concentration and thereafter dialyzed overnight against distilled water. This substance and pI markers (Pharmacia Biotec Broad pI calibration kit, pH 3.5–9.3) were subjected to an isoelectric focusing gel (Novex pH 3–10) that ran at 100 V for 60 min, 200 V for 60 min, and 500 V for 30 min under native conditions. The gel was fixed in TCA (12 g), and sulfosalicylic acid (3.5 g) was added to 100 ml deionized water and stained with Colloidal blue (Novex).

Sampling of tumor venous blood. To examine whether the UV peak around 40 kDa (see *Characterization of fluid isolated from tumors and subcutis*) was present in venous blood draining the tumor, blood was sampled with micropipettes from surface veins of five tumors in three rats. After anesthesia, the tumor surface was exposed by a skin incision. Under guidance of a Zeiss stereomicroscope (magnification ×40), a micropipette with tip diameter ~50 μm was inserted into a venule draining the tumor. After aspiration of 5–20 μl of blood, the pipette was withdrawn, and the blood was dissolved in 0.5 ml of HPLC buffer. This sample was centrifuged, and the red blood cell-free supernatant was processed for elution in the HPLC column.

Composition of the tumor extracellular matrix. As shown in previous publications, the composition of the extracellular matrix may influence the diffusion properties of the tumor interstitium (14) and the composition of the fluid isolated by centrifugation (1, 3). Therefore, the tumor content of collagen, hyaluronan, and total glycosaminoglycans (assayed as uronic acid) was measured. Analysis of hyaluronan was performed using a radioassay (HA-test 50, Pharmacia Diagnostics; Uppsala, Sweden) after papain digestion of freeze-dried specimens (16).

Uronic acid was measured by the method of Bitter and Muir (5) as described in a previous paper (15), whereas collagen was determined according to the method of Woessner (27), based on the determination of hydroxyproline content as described in a previous publication (15) assuming a hydroxyproline content of 0.91 μmol/1 mg collagen (4).

Statistics. Data are given as means ± SE. Data from consecutive centrifugations were compared using one-way ANOVA and with Tukey tests for multiple comparisons. Differences were accepted as statistically significant at the $P < 0.05$% level.

RESULTS

The centrifugal force needed to isolate any fluid from the tumors varied. In some, 500 rpm (27 *g*) was sufficient, whereas 800 rpm (68 *g*) was sufficient for fluid isolation in all tumors. The fluid isolated was straw colored, and sometimes some erythrocytes collected at the bottom of the centrifugation tube, most often in the initial centrifugation (27–68 *g*) of the consecutive centrifugations. Fluid isolated from the central tumor was more grayish than that of the other tissues and became more viscous at increasing G force.

The weight fractions of fluid isolated from the whole tumor ($n = 19$), tumor pole ($n = 8$), central tumor ($n = 5$), and skin ($n = 9$) are shown in Fig. 1. In the initial of the consecutive centrifugations (27–68 *g*), the isolated fractions ranged from 0.013–0.017, with a corresponding range of 0.030–0.043 at 106 *g*. At 424 *g*, the accumulated fraction isolated from the central tumor of 0.10 differed significantly from those derived from the other types of tissue (range 0.050–0.069). At even higher centrifugation speeds, the isolated fractions from tumor tissue differed significantly from that of the skin, e.g., with a fraction of 0.191 for central tumor and 0.089 for skin at 6,800 *g*.

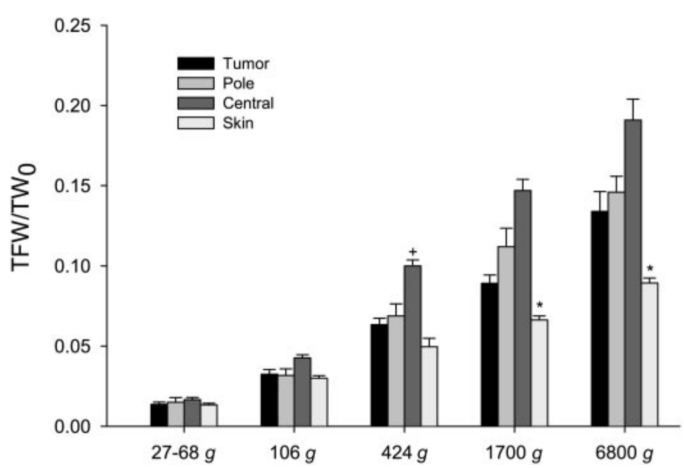

Fig. 1. Cumulative tissue fluid weight as a fraction of initial sample weight ($\Sigma TFW/\Sigma TW_0$) as a function of G force for intact tumor (tumor), tumor pole (pole), excised central tumor (central), and skin (means ± SE). +Significantly different ($P < 0.05$) from intact tumor and skin; *$P < 0.05$ compared with all types of tumor tissue.

Analyses of tumor fluid. In these experiments we measured recovered tracer that had equilibrated in extracellular fluid (^{51}Cr-EDTA) or distributed in plasma (^{125}I-labeled human serum albumin), and the results are shown in Fig. 2. For the intact, whole tumor, the concentration of ^{51}Cr-EDTA in tumor fluid relative to plasma averaged 0.922 (± 0.035), 0.952 (± 0.028), and 0.987 (± 0.028) in consecutive centrifugation at 27–68, 106, and 424 *g*, respectively (Fig. 2*A*). None of these ratios differed significantly from 1.0. The corresponding numbers for the tumor pole were 0.830 (± 0.019), 0.966 (± 0.034), and 0.954 (± 0.063). Of these ratios, only that obtained at the lowest G force differed significantly from 1.0 ($P < 0.001$). The ^{51}Cr-EDTA central tumor-to-plasma ratio ranged 0.67–0.77 for the three lowest centrifugation speeds, all significantly <1.0 ($P < 0.001$ for all comparisons).

Increasing the G force to 1,700 and 6,800 *g* (4,000 and 8,000 rpm, respectively) resulted in a gradual fall of the ^{51}Cr-EDTA tissue-to-plasma ratio, e.g., in whole tumor to 0.882 (± 0.020) and 0.772 (± 0.043). All ratios obtained at these centrifugation speeds were significantly different from 1.0.

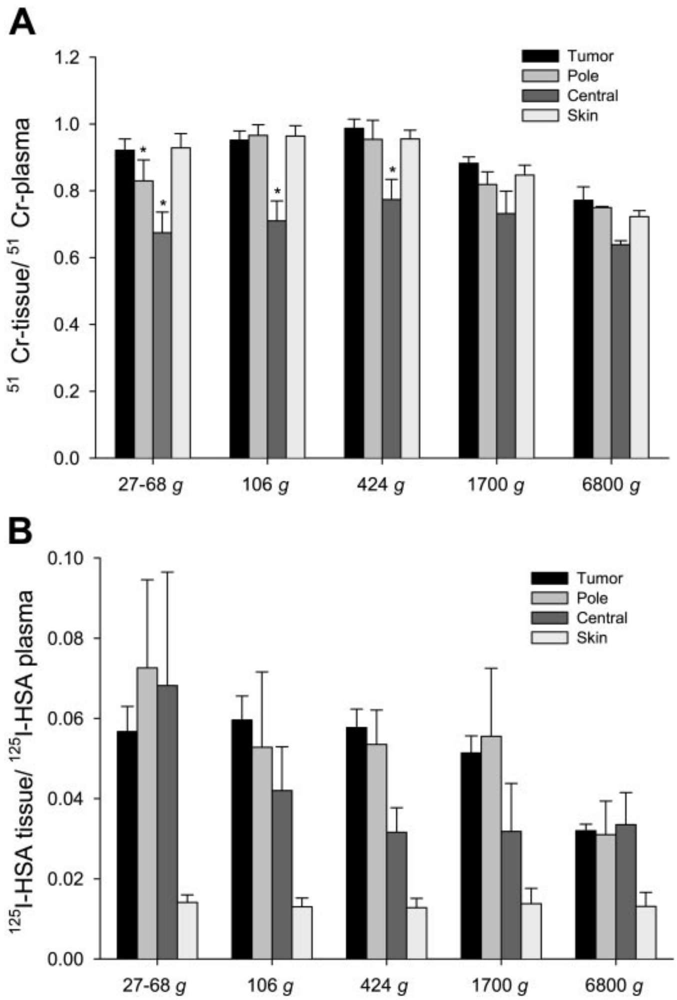

Fig. 2. Tissue-plasma distribution of the extracellular tracer ^{51}Cr-EDTA (*A*) and intracellular tracer ^{125}I-human serum albumin (HSA) (*B*) as a function of G force for intact tumor (tumor), tumor pole (pole), excised central tumor (central), and skin (means ± SE). *Significantly different ($P < 0.001$) from 1.0. For 1,700 and 6,800 *g*, the ^{51}Cr-EDTA ratios for all tissues were significantly different from 1.0.

In the skin, the mean tissue-to-plasma ratio ranged from 0.93–0.96 for G ≤ 424; none of the ratios were significantly different from 1.0 (Fig. 2*A*). Increasing the G force in the consecutive centrifugations to 1,700 and 6,800 *g* resulted in a corresponding ratio of 0.85 and 0.72, respectively, both significantly different from 1.0 ($P < 0.001$).

The results from the ^{125}I-labeled human serum albumin recovery in centrifuged fluid are shown in Fig. 2*B*. For the intact tumor, the tracer concentration relative to plasma was 0.05–0.06 for the first four consecutive centrifugations (≤1,700 *g*), falling to 0.032 at the highest G force (6,800 *g*). The corresponding average ratios for the pole and central tumor differed slightly but not significantly from that of the whole tumor for all centrifugation speeds. In fluid isolated from the skin, the ^{125}I-labeled human serum albumin ratio ranged from 0.0130–0.0142, all significantly different from the ratio of fluid isolated from tumor at all centrifugation speeds.

COP. The COPs in fluid isolated in subsequent centrifugations from the tumor and skin are shown in Fig. 3. In the intact tumor, the average pressure ranged from 14.4–15.9 mmHg for 27–68 to 1,700 *g* and was 17.7 mmHg at the highest G force induced by 6,800 *g* (Fig. 3*A*). The corresponding pressure in plasma was 20.5 mmHg (± 0.8, n = 14). All pressures except for that obtained at 6,800 *g* was significantly different from the COP in plasma. No data were obtained for the tumor pole at the lowest and highest centrifugation speed, but for 106–1,700 *g*, the COP in the fluid was similar to that of the whole tumor.

When assayed on the colloid osmometer, fluid isolated from the central tumor showed a different pattern from that of the whole tumor and tumor pole. Whereas recordings from the two latter became stable within 1–2 min, fluid from the central tumor initially had a higher pressure that gradually fell off to a stable level, indicating passage of molecules through the osmometer membrane. The difference between the initial and stable pressure increased with increasing G force. Thus, at 106 *g*, the initial pressure averaged 17.1 mmHg (± 2.1, n = 12) (Fig. 3*A*), whereas the stable pressure was 13.7 (± 1.8, n = 12). The corresponding pressures obtained at 1,700 *g* were 32.1 (± 2.1, n = 11) and 18.3 mmHg (± 3.6, n = 11).

COP in fluid isolated from the skin was generally lower than that of the tumor. The average pressure ranged from 11.4 mmHg (± 1.4, n = 8) (obtained at 6,800 *g*) to 13.8 mmHg (± 1.0, n = 8) (at 424 *g*), with a corresponding COP in plasma of 23.0 mmHg (± 0.3, n = 8) (Fig. 3*A*). Pressures in the skin did not differ significantly from each other, and when compared with the COPs in whole tumor and tumor pole only, the pressures in fluid isolated in whole tumor and skin at 6,800 *g* differed significantly.

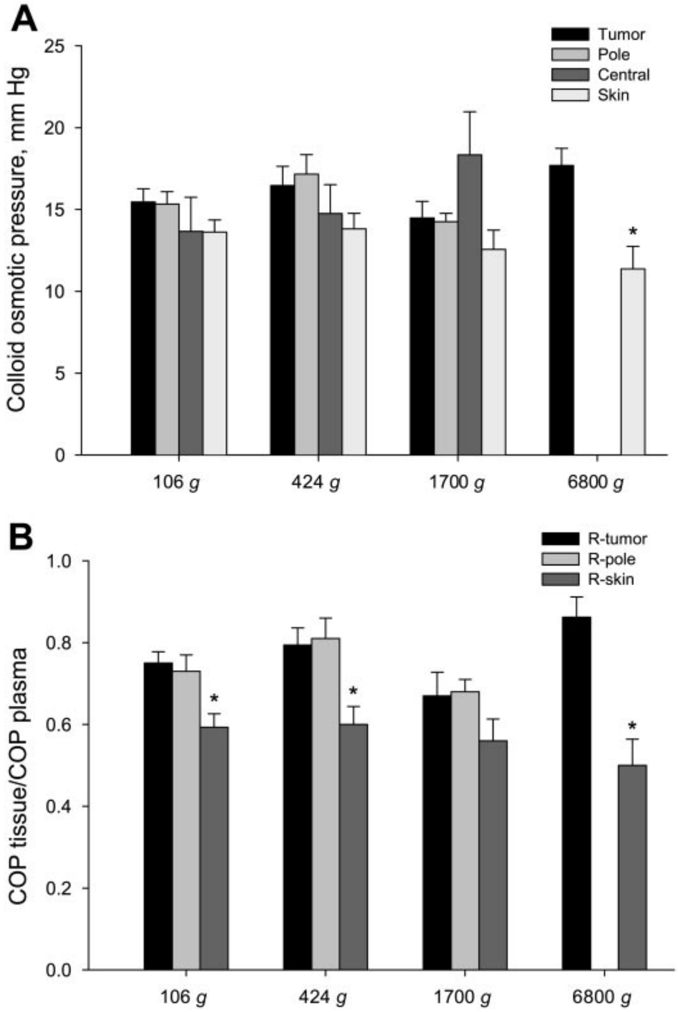

Fig. 3. *A*: colloid osmotic pressure (COP) as a function of G force for intact tumor (tumor), tumor pole (pole), excised central tumor (central), and skin (means ± SE). *Significantly different ($P < 0.05$) from intact tumor at same G force. *B*: COP in tissue fluid relative to plasma as a function of G force for intact tumor (R tumor), tumor pole (R pole), and skin (R skin) (means ± SE). *$P < 0.05$ compared with both types of tumor tissue. Numbers for central tumor not included because of pressure transients (see text).

The level of plasma COP may influence tissue fluid COP, and therefore the tissue-to-plasma COP ratios were calculated. As displayed in Fig. 3*B*, this ratio was 0.75, 0.79, 0.67, and 0.86 in intact tumor ($n = 14$ for all ratios) at 106, 424, 1,700 and 6,800 *g*, respectively. The corresponding ratios for the skin were 0.59, 0.60, 0.55, and 0.50 ($n = 8$ for all ratios). All corresponding ratios in the tumor and skin, except that obtained at 1,700 *g*, differed significantly.

In tumor pole fluid, the COP ratios to plasma were 0.73, 0.81, and 0.68 ($n = 6$ for all) for 106, 424, and 1,700 *g*, respectively. Of these, the two former ratios differed significantly from the corresponding ratios for skin. Because of the difficulties establishing stable COP for central tumor, no ratios were calculated for this tissue.

Analyses of isolated fluid. The elution pattern of fluid isolated from an intact tumor by centrifugation at 106 *g* is shown in Fig. 4*B*, also showing the elution pattern of plasma for comparison (Fig. 4*A*). We observe that for albumin and molecules larger than albumin, the elution pattern is similar for the globulins and albumin. For molecules smaller than albumin, the elution pattern of tumor fluid and plasma differed significantly. In tumors, a macromolecule with molecular weight less than albumin consistently eluted at 13.4 ml, and such a macromolecule was never seen in the arterial plasma samples. The area of this peak ranged from 8.9% to 32.1% (average 16.3%) of the area under the elution curve when measured in 13 tumors centrifuged at 27–6,800 *g* and did not change with centrifugation speed. Furthermore, the relative areas for intact tumor and tumor pole were similar. In addition, there were some even smaller macromolecules in tumor fluid that were or were not found in plasma, constituting a variable but small fraction (<8.3%) of the total area under the elution curve.

The pattern for the central tumor, however, differed significantly from the intact tumor and tumor pole (Fig. 4*C*). As for the other tumor eluates, the specific tumor peak was found also for the central tumor. In addition, several smaller molecular weight fractions were found constituting a larger area than for intact tumor and tumor pole (up to 45%), most likely representing intracellular proteins.

Fluid was also isolated from the skin by centrifugation and by wick implantation, and examples of elution patterns for these fluids using the Superose 12 column are shown in Fig. 4, *D* and *E*, respectively. The elution pattern for wick fluid was similar to that of plasma except for a relatively large albumin fraction. Fluid isolated from the skin by centrifugation was practically identical to wick fluid, except for a protein eluting at 13.4 ml observed in some samples. However, this constituted <5% of the total area under the curve.

The elution pattern of plasma collected by aspiration of blood by a micropipette at the tumor surface is shown in Fig. 4*F*. We observe that this pattern is similar to that of the tumor centrifugate and particularly note that the specific tumor macromolecule smaller than albumin was present in venular plasma.

To quantify the amounts of albumin and globulin in centrifugates from intact tumors, standard HPLC curves were made from known amounts of albumin. By comparing these to curves from plasma where total protein concentration was determined by refractometry enabled us to measure albumin concentration in plasma and globulin as the difference between total protein and albumin concentration. Results from these calculations are shown in Table 1. We observe that for G force ≤424, the average ratio between albumin in centrifugate and plasma ranged from 0.70–0.76, with a corresponding ratio for globulins of 0.70–1.06.

We wanted to compare fluid isolated from tumors by centrifugation to that eluted into a buffer to see whether the centrifugation process per se affected the composition of the fluid. Tumors were soaked in PBS for 24 and 48 h, and tumor eluate was isolated. In addition, we isolated extracts from tumors where cell

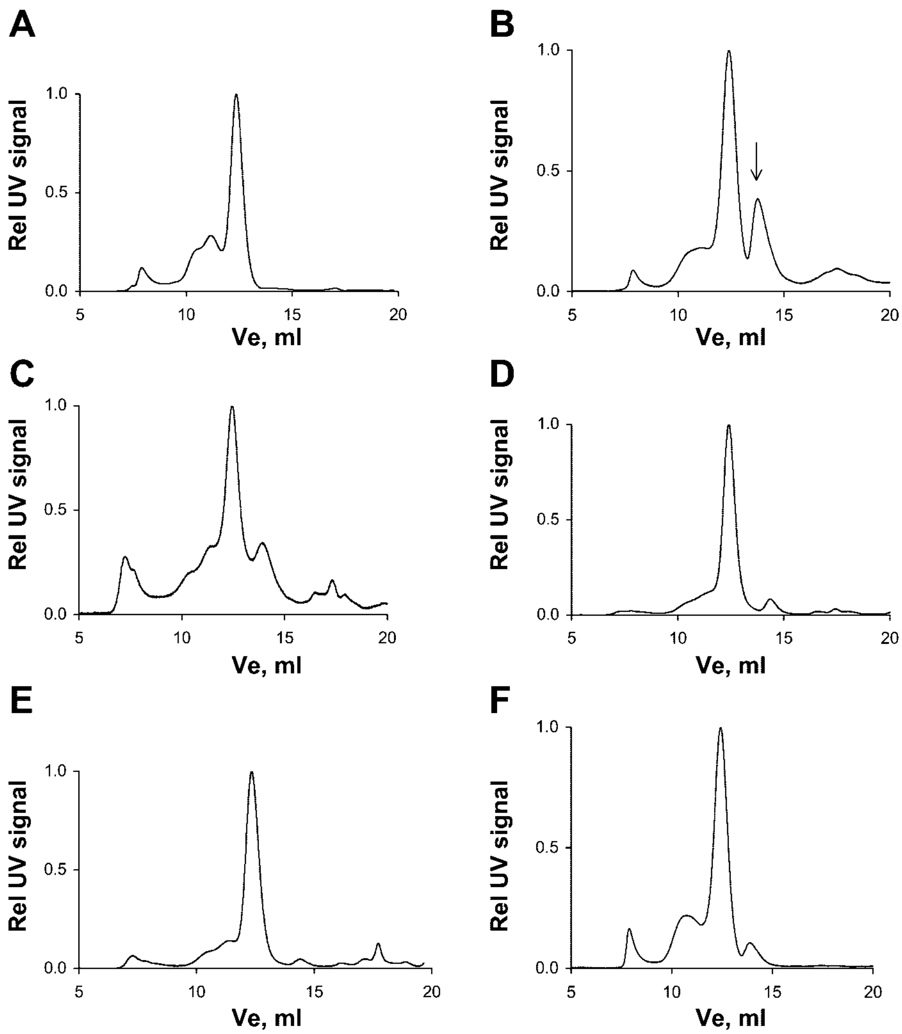

Fig. 4. Patterns for plasma and tissue fluid samples eluted (V_e) in Superose 12 HR column. *A*: plasma; *B*: centrifugate (106 *g*) from intact tumor; *C*: centrifugate (106 *g*) from central tumor; *D*: centrifugate (106 *g*) from back skin; *E*: wick fluid from back skin; *F*: venous plasma from tumor. Arrow, fraction of extracellular macromolecule prominent in tumor fluid.

damage had been induced. These tumors were freeze-dried and thereafter crushed by using a dismembranator before being soaked in PBS for 24 h. Fluid isolated this way from intact and crushed tumor tissue was passed through the Superdex 75 column, giving a better separation in the small molecular weight range than the Superose 12 column. Fluid from the intact tumor (Fig. 5*B*) had an elution pattern resembling that of plasma (Fig. 5*A*), except for a marked peak at 13.4 ml, corresponding to that observed in centrifugate from intact tumor. Elution patterns for fluid used for soaking the crushed tumor differed substantially from plasma and intact tumor (Fig. 5*C*), with several peaks in the area where macromolecules with less molecular weight than that of albumin elutes.

Table 1 *Concentration of albumin and globulins in tumor fluid isolated by centrifugation and plasma*

	Albumin, mg/ml	Globulins, mg/ml
Plasma	44.4 ± 0.62(8)	10.5 ± 0.33(8)
≤800 *g*	31.2 ± 3.5(6)	7.3 ± 0.72(6)
106 *g*	30.9 ± 2.2(8)	7.5 ± 1.1(8)
424 *g*	33.5 ± 3.3(8)	11.2 ± 1.6(8)
1,700 *g*	28.1 ± 2.4(9)	8.4 ± 1.2(9)

Values are means ± SE; number of observations are in parentheses.

Plasma, tumor centrifugates, and eluate from intact and crushed tumor were also used for SDS electrophoresis. The pattern reflected that obtained by HPLC, but with less resolution, and therefore the data have not been shown.

Electrophoresis of the fraction isolated from tumor eluate at 13.4 ml showed at least three distinct bands. Although unspecific, this observation suggests that several typical tumor macromolecules not found in plasma by HPLC are present in the tumor interstitium.

Fluid volumes and composition. Tumors ($n = 7$) had a water content of 0.83 ml/g (± 0.002). Hyaluronan and uronic acid in tumors averaged 1.92 (± 0.26) and 1.47 (±0.43) mg/g wet wt, respectively, whereas the mean collagen content was 4.6 mg/g wet wt (± 0.7).

The extracellular and intravascular fluid distribution volumes for intact tumors ($n = 11$) were 0.40 ± 0.02 and 0.014 ± 0.001 ml/g, respectively, with corresponding volumes in the back skin ($n = 7$) of 0.46 ± 0.03 and 0.005 ± 0.001 ml/g.

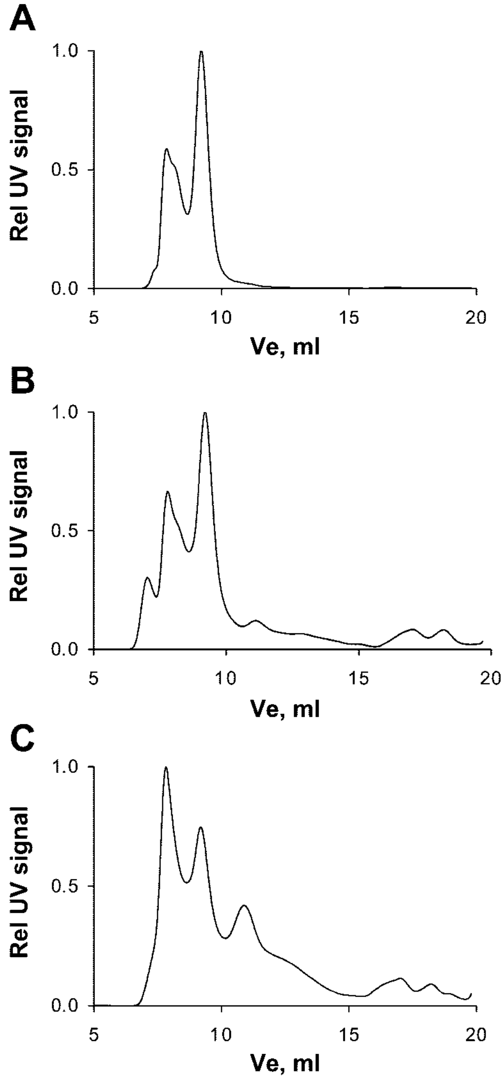

Fig. 5. Patterns for plasma and fluid extracted by tissue elution passed through a Superdex 75 HR column. *A*: plasma; *B*: eluate from intact tumor eluted in PBS for 24 h; *C*: eluate from crushed tumor.

The tracer injected could be totally eluted from the tissue. Thus after elution, <1% of the initial counts of ^{51}Cr-EDTA (range 0.40–0.96%) remained in the tumor ($n = 5$) and skin ($n = 4$). The corresponding range for ^{125}I-labeled human serum albumin was 0.1–2.3%.

DISCUSSION

We have shown that it is possible to isolate tumor fluid even at low-speed centrifugation. We have evaluated the isolated fluid, and the major conclusion of our work is that when centrifuging intact tumor or tumor pole at ≤424 *g* (here 2,000 rpm), it is possible to isolate fluid that is representative for tumor interstitial fluid. At higher G force, dilution of the centrifugate occurs most likely by intracellular fluid. Centrifuging excised central tumor tissue results in a centrifugate where the extracellular tracer is diluted and contaminated by intracellular proteins and/or tumor degradation products. In the following, we discuss the basis for our conclusion, limitations of the method, and implications for further work. In agreement with previous studies, this fluid had a high protein concentration and COP (8, 19). Furthermore, our experiments suggest that typical tumor macromolecules, most likely proteins, are liberated to the interstitial fluid and general circulation, in agreement with previous work (21).

Evaluation of the centrifugation method. In previous studies, we used centrifugation to isolate fluid from the cornea (22) and tail tendon (3), both being tissues with a low cell content. Tumors are rich in cells, and cell compression during centrifugation may lead to extrusion of cellular fluid, resulting in the isolation of a mixture of interstitial and cell fluid. We decided to use ^{51}Cr-EDTA as a probe to show possible "contamination" of cellular fluid. ^{51}Cr-EDTA (molecular weight 341) is a substance that is not metabolized and is not taken up by cells (12). Addition of tumor cell fluid to the centrifuged volume should thus show up as a reduced ^{51}Cr-EDTA concentration in the centrifugate relative to plasma. For tissue centrifugation at a G force ≤424, we found that the ratio of ^{51}Cr-EDTA in centrifugate and plasma was not significantly different from 1.0, suggesting no dilution of extracellular fluid. Furthermore, the unaltered COP with increasing G force up to 424 suggests no sieving of macromolecules in this G force range. The observed gradual reduction of this ratio when exposing the tissue to higher G forces indicates that a gradual dilution is occurring, which may be a result of extrusion of cellular fluid and/or cell damage. The somewhat lower ^{51}Cr-EDTA centrifugate-to-plasma ratio of in the initial sample isolated at 27–68 *g* may be a result of flushing of the tumor with saline to remove surface blood.

By centrifugation, we sample from the whole extracellular phase, including the vascular volume. To quantify the latter fraction, we injected an intravascular tracer and found a concentration in centrifugate 3–6% of that in plasma (Fig. 2*B*). This fraction may be indicative of the size of the fluid fraction originating intravascularly, but because of the increased "leakyness" of tumor capillaries to proteins (9), it may represent an overestimate of the true intravascular volume fraction. The significantly lower intravascular volume determined with the same tracer in whole tumor (1.4 ml/100 g, cf, *Fluid volumes and composition*) may seem puzzling. A likely explanation for this finding is that the determination of intravascular volume represents an average for all regions in the tumor, whereas centrifuged fluid derives mainly from the well-vascularized tumor periphery.

Another potential problem associated with our method might be cell damage during centrifugation resulting in contamination of the sample with intracellular proteins. In the initial phase of the study, we suspected that such contamination occurred judging from the HPLC of tumor fluid showing a peak in tumor fluid not present in plasma. To address this question we passed through the HPLC column extracts from homogenized tumors and fluid used for elution of whole tumors. Whereas tumor eluate had a HPLC pattern similar to that of plasma, except for the fraction with lower molecular weight than albumin, i.e., similar to tumor centrifugate, tumor homogenate had a much higher fraction of low-molecular-weight macromolecules. This is to be expected because a large fraction of the latter consists of intracellular proteins. The final evidence that the observed macromolecule(s) was present in tumor interstitial fluid came from the demonstration of the same peak in tumor venular plasma. This peak was present in all blood samples isolated, constituting ~5% of the total amount of protein. Apart from showing its presence in the interstitium, this observation also shows that a typical tumor macromolecule(s) (molecular weight ~40,000) must be produced at a considerable rate to become detectable in such high quantity in venous plasma. Although the exact nature of the substance is not established, these data support the observations of Sylven and Bois (21), showing leakage of intracellular enzymes (peptidases, molecular weight ~40,000) to interstitial fluid.

As mentioned in the introduction, tumor tissue should be well suited for isolation of interstitial fluid by centrifugation because of the high hydraulic conductivity (9, 20). However, substances constituting the extracellular matrix, i.e., collagen and glycosaminoglycans, may impede the filtration of extracellular fluid from the tumor during centrifugation. Furthermore, intrafibrillar water may be extruded from collagen fibrils, as shown in experiments in the cornea (22) and tail tendon (3), resulting in an increased dilution of plasma proteins with increasing G force. Compaction of the extracellular components may also lead to formation of a "condensation-polarization" boundary layer (17), which may also contribute to sieving of macromolecules in the interstitium. To this end, we measured extracellular matrix components in this tumor model and verified that collagen and glycosaminoglycans were low compared with skin, in agreement with previous studies in other tumors (6, 9, 13). Unlike centrifugation experiments in other tissues, we found no evidence of sieving and dilution of macromolecules in tumor by exposing the tissue to ≤ 424 *g* (2,000 rpm), because the protein concentration was unrelated to the G force in this range, a finding that most likely is a result of the composition of the extracellular matrix of the tumor. Centrifugation at 1,700 and 6,800 *g* resulted in dilution of the centrifugate by fluid from a space not accessible to ^{51}Cr-EDTA, most likely the intracellular phase.

Comparison to other available methods. Few other methods have been described for isolation of interstitial fluid. Sylven and Bois (21) created pouches in central and peripheral tumor tissue by blunt dissection and sampled fluid that collected in these by micropipettes. As argued by the authors, the fluid from the periphery was not from tumor only but also from surrounding normal tissues. The fluid from the tumor center was not mixed but included necrotic tissue and may therefore not be representative for interstitial fluid. Another approach was used by Gullino and coworkers (8). They created a chamber within a tumor separated from the tumor tissue by a porous membrane. Fluid draining from the tumor into the chamber could be sampled with a catheter. The advantage of that procedure will be the possibility to sample for longer periods and that fluid is easily obtainable, but capsule implantation may result in scarring around the sampling device, i.e., a connective tissue not representative for tumor tissue. In a recent study published from Jain's group (19) wicks were used to obtain interstitial fluid. Wicks were either inserted simultaneously with the tumors (chronic wicks) or acutely at the termination of the experiment. As pointed out by the authors, one problem with the use of this method in the highly vascular tumor tissue is blood contamination of the isolated wick fluid. Even though the authors claimed that this problem was avoided by cutting off blood-stained wick portions, our experience is that blood contamination is difficult to avoid in tumors. Another difficulty that may be faced when using the wick technique in cell-rich tumor tissue is contamination of wick fluid with intracellular proteins, i.e., mainly proteins with lower molecular weight than albumin, as observed in skeletal muscle (25). These reservations have to be considered when discussing data on tumor interstitial fluid isolated using wicks.

Functional importance of the data. The fluid isolated from tumors by centrifugation had a COP of ~75% of that in plasma, which was higher than the corresponding ratio of 50% found in skin, showing a high protein concentration in the tumor interstitium. Using the implanted chamber technique discussed above in rats, Gullino et al. (8) found an average tumor interstitial fluid protein concentration, 67% of that in plasma, whereas the corresponding ratio for subcutis was ~50%. In mammary carcinomas in mice, Sylven and Bois (21) observed a tumor fluid protein concentration not different from that in plasma, in agreement with the work of Stohrer et al. (19), who found a COP in wick fluid of three of the four tumor types not significantly different from that in plasma. Although there are differences in the absolute numbers, that partly may be related to tumor types, species, and the methods used for sampling, previous as well as the present data suggest that the protein concentration and COP in interstitial fluid of tumors are higher than in normal tissue. This is most likely a consequence of a low size selectivity of tumor capillaries (10, 28) and the lack of functional lymphatics in tumor tissue (11). The functional importance of the low COP gradient across the tumor capillary will be less than in normal tissue for determining net filtration pressure and thereby transcapillary fluid exchange, which will then mainly be governed by the difference in hydrostatic pressure in capillary and interstitial fluid.

In conclusion, we developed a centrifugation method to isolate interstitial fluid from tumors. Centrifugation of tumor tissue at $\leq$424 *g* resulted in the isolation of fluid with a similar concentration of the extracellular tracer ^{51}Cr-EDTA as in plasma. Our data suggest that typical tumor macromolecules not found in plasma are dissolved in the interstitial fluid and that the tumor interstitial fluid COP is higher than in skin. Even though we have evaluated the procedure in one tumor type only, our data also suggest that low-speed tumor centrifugation may be a simple and reliable method when studying interstitial transcapillary fluid balance and interstitial distribution of probes, including therapeutic agents.

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